

Fascinating facts

IN NUMBERS

- The largest stable nucleus belongs to **lead** – it has **208 subatomic particles** (126 neutrons and 82 protons).

- **Bismuth** is very slightly radioactive. It would take **20 million trillion years** for half of its atoms to decay. At less than 14 billion years of age, even the Universe is not that old!

- **Buckminsterfullerene** is one of the allotropes (physical forms) of carbon. One molecule of it is made of **60 atoms of carbon** arranged in a ball.

- The **body** of an average adult human contains **43 kg** (94.79 lb) of oxygen and **0.2 mg** (0.0000071 oz) of gold.

- Around **1.1 billion tonnes** (1.2 billion tons) of **iron** are produced annually – enough to make a cube with sides about 720 m (2,362 ft) long, which is more than twice the height of the **Eiffel Tower** in Paris, France.

- Neutron stars are made only of neutrons and not atoms. Astronomers have given this material the nickname **neutronium**. One teaspoon of it would weigh around **90,718,000 tonnes** (100 million tons)!

BREAKING THE BANK

- The **Large Hadron Collider**, the world's largest particle accelerator, has around **US\$120 million** worth of wire, made from the metals niobium and titanium, in it.

- In the 1850s, **aluminium** was twice as expensive as **gold** – it was even used to make dinner plates for royalty.

- **Palladium** is the most expensive precious metal. It costs almost **100 times** as much as the **same quantity of silver**.



The aluminium foil used in kitchens is 16 millionths of a metre (0.00063 in) thick.

WEIRD BUT TRUE

- **Mercury** is a liquid that is **13.5 times denser** than water. A person could sit on top of a pool of mercury without sinking.

- Recycling **aluminium** uses just **5 per cent** of the energy needed to extract pure uranium from its ores.

- **Lightning strikes** occur nearly **50 per cent** more often above nuclear power plants because the reactors release charged **krypton** gas.